

Title of Your Notes

Your Name

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1. Theorem Environments

§1.1. Main Environments

Prototypical example for this section: The integers $\mathbb{Z}$ with addition.

Definition 1.1.1. A **group** is a set G together with a binary operation $\cdot$ satisfying associativity, identity, and inverses.

Theorem 1.1.2

Every subgroup of a cyclic group is cyclic.

Proof. Let $G = \langle g \rangle$ and $H \leq G$. Take the smallest positive n with $g^n \in H$; then $H = \langle g^n \rangle$. $\square$

Lemma 1.1.3

The intersection of two subgroups is a subgroup.

Corollary 1.1.4

The intersection of any collection of subgroups is a subgroup.

Proposition 1.1.5

Every group of prime order is cyclic.

§1.2. Remarks and Examples

§1.2.i. Examples

Example 1.2.1

The even integers $2\mathbb{Z}$ form a subgroup of $(\mathbb{Z}, +)$.

Remark 1.2.2 — Note that $\mathbb{Q}$ under multiplication is *not* a group because 0 has no inverse.

A group is the minimal algebraic structure in which “symmetry” makes sense.

§1.2.ii. Other Environments

Claim 1.2.3. The center $Z(G)$ is always a normal subgroup of G .

Proof. For any $z \in Z(G)$ and $g \in G$, we have $gzg^{-1} = gg^{-1}z = z \in Z(G)$. ■

Fact 1.2.4. Every finite group of odd order is solvable (Feit–Thompson).

Numbered vs unnumbered

Use `theorem` for numbered environments and `theorem*` for unnumbered ones (not defined in Napkin by default; add manually if needed).

§1.3. Exercises

Exercise 1.3.1. Show that $(\mathbb{Z}/p\mathbb{Z})^\times$ is a group under multiplication for any prime p .

Solution. Since p is prime, every nonzero element of $\mathbb{Z}/p\mathbb{Z}$ has a multiplicative inverse by Bezout’s theorem. □

2. Diagrams and Macros

§2.1. Commutative Diagrams

Example 2.1.1

A commutative square:

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & & \downarrow h \\ C & \xrightarrow{k} & D \end{array}$$

§2.2. Useful Math Macros

§2.2.i. Arrows

- Injection: $f: A \hookrightarrow B$ (`\injto`)
- Surjection: $f: A \twoheadrightarrow B$ (`\surjto`)
- Labeled: $A \xrightarrow{f} B$ (`\taking{f}`)

§2.2.ii. Shorthands

- $\overline{AB}$, $\tilde{X}$, $\hat{f}$ (`\ol`, `\wt`, `\wh`)
- $x := y$ (`\defeq`)
- ε , $^{-1}$ (`\eps`, `\inv`)
- $\lfloor x \rfloor$, $\lceil x \rceil$, $|x|$, $\|x\|$

§2.2.iii. Number Fields

$\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{F}$, $\mathbb{E}$